7. Western Blotting technique
8. Neutralization test

### 1. Precipitation reactions

When a soluble antigen combines with its antibody in the presence of electrolytes (NaCl) at a suitable temperature and pH, the antigen-antibody complex, forms an insoluble (visible) precipitate and this reaction is called precipitation. When instead of sedimenting, the precipitate remains suspended as floccules, the reaction is known as flocculation.

- **Applications of precipitation reactions**

The following types of precipitation tests are in common use:

**a) Ring test**

Examples of ring precipitation test are the C- reactive protein test, Ascoli's thermoprecipitin and the grouping of streptococci by the Lancefield technique.

**b) Slide test**

When a drop of antigen and a drop of antiserum are placed on a slide and mixed by shaking, floccules appear. The VDRL test for syphilis is an example of slide flocculation.

**c) Tube test**

A quantitative tube flocculation test is used for the standardization of toxins and toxoids.
Precipitation reaction in gels

There are several advantages in allowing precipitation to occur in a gel rather than in a liquid medium. The reaction is visible as a distinct band of precipitation, which is stable and can be stained for preservation, if necessary. Imunodiffusion is usually performed in 1% agarose gel. Different modifications of the test are available.

- Single Diffusion in One Dimension (Oudin Procedure)
- Double Diffusion in One Dimensions (Oakley-Fulthorpe Procedure)
- Single Diffusion in Two Dimensions (Mancini Procedure)
- Double Diffusion in Two Dimensions (Ouchterlony Procedure)

**Immunoelectrophoresis**

Immunoelectrophoresis was devised by Grabar and Williams (1953). This method consists of two steps. The first step is agarose electrophoresis of the antigen. Rectangular trough is then cut into the agarose gel parallel to the direction of the electric field and is filled with the antiserum. By diffusion, lines of precipitation develop with each of the separated compounds (Figure 13.24). This method is used to detect normal and abnormal serum proteins.

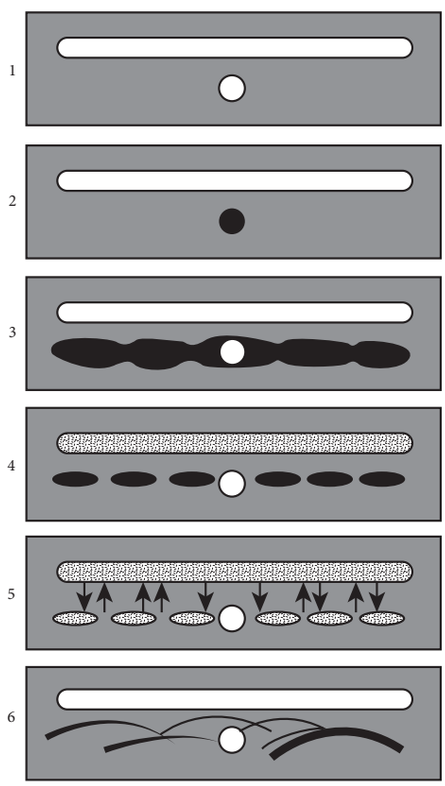


**Figure 13.28:** Immunoelectrophoresis



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